

Computer Science student with hands-on experience in full-stack development, systems programming, and machine learning. Skilled in building scalable applications, implementing system-level solutions, and analyzing data to extract meaningful insights. Strong foundation in algorithms, data structures, and databases, with practical experience across backend development, desktop applications, and AI models.

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2022 – Present

GPA: 3.4 | Languages: Arabic (Native), English (Very Good)

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C++, Python, JavaScript, SQL, C#

React.js, Node.js, Express.js, NestJS, .NET, MongoDB, RESTful APIs, JWT Authentication

Data Analysis, Data Preprocessing, Scikit-learn, Model Evaluation, TF-IDF, Bag of Words

Operating Systems (Memory Management, CPU Scheduling), Parallel Programming (MPI)

Git, GitHub, VS Code, Visual Studio, Oracle SQL, Troubleshooting

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— MERN Stack (Team Project)

- Collaborated within a team to build a full-stack LMS with role-based access control for Admins, Instructors, and Students, using React.js, Node.js, Express.js, and MongoDB.
- Implemented JWT authentication and authorization, plus an instructor registration and admin approval workflow.
- Delivered 6 core features: course creation & publishing, rejection/resubmission workflow, enrollment, progress tracking, certificate generation, and coupon/discount management.
- Added search & filtering and complete CRUD for courses, sections, and lessons.

- Built a machine learning model using Python and Scikit-learn, applying BoW and TF-IDF techniques.
- Trained and compared Naive Bayes, Logistic Regression, and SVM, achieving 99.38% accuracy with SVM.

— C# & Oracle SQL

- Developed a secure transaction system handling authentication, login, and money transfers.

— Memory Management & CPU Scheduling ([GitHub](#))

- Implemented dynamic memory allocation (First Fit, Best Fit, Worst Fit, Next Fit) with segregated free lists and block splitting/coalescing.
- Built page-level memory management, tracking block sizes and reclaiming fully free pages back to the system.
- Implemented CPU scheduling enhancements: priority management, dynamic ready-queue placement, and starvation handling via timer-based logic.

— C++ & Qt ([LinkedIn](#))

- Built a desktop wallet application with an interactive GUI using C++ and the Qt framework, applying data structures for efficient data handling.
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— DEPI (Digital Egypt Pioneers Initiative)

Jul 2026 – Present

- Accepted into DEPI's competitive Full Stack .NET Web Development track, a national technical training initiative.
- Building full-stack web applications through hands-on frontend and backend training using .NET and modern web development practices.
- Collaborating with peers on real-world development tasks, applying Git-based version control and adapting to new tools and workflows.

— ITI (Information Technology Institute)

Jul 2026 – Present

- Training in backend development with Node.js, building hands-on experience with each new module through applied session work.
- Working with Express.js and NestJS to build and structure RESTful APIs, alongside advanced JavaScript beyond core fundamentals.

— Oil & Gas Solutions Company

- Performed data cleaning, preprocessing, and analysis on company databases, communicating findings clearly to support decision-making.

— UNESCO Center for Peace Camp

- Led and facilitated STEM-related activities and workshops for program participants.
- Guided participants through collaborative problem-solving tasks, building leadership and mentoring skills.

— Ain Shams University

- Built responsive and user-friendly web interfaces using HTML and JavaScript, managing time across multiple concurrent training tasks.
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- Soft Skills Training Program — strengthened communication, teamwork, and presentation abilities through structured group discussions and collaborative projects.